

Adaptive Rejection of Unknown Hybrid Sinusoidal Disturbances for Bipedal Walking on Multi-Directional Moving Platforms

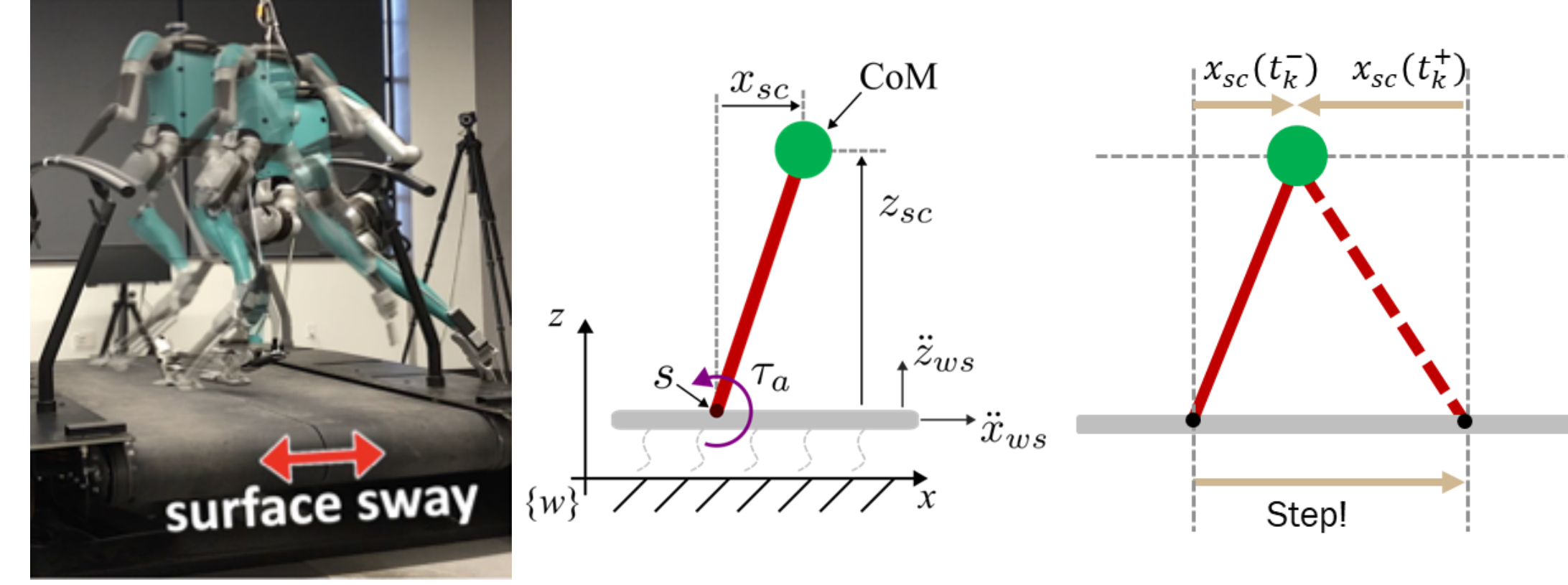
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OVERVIEW

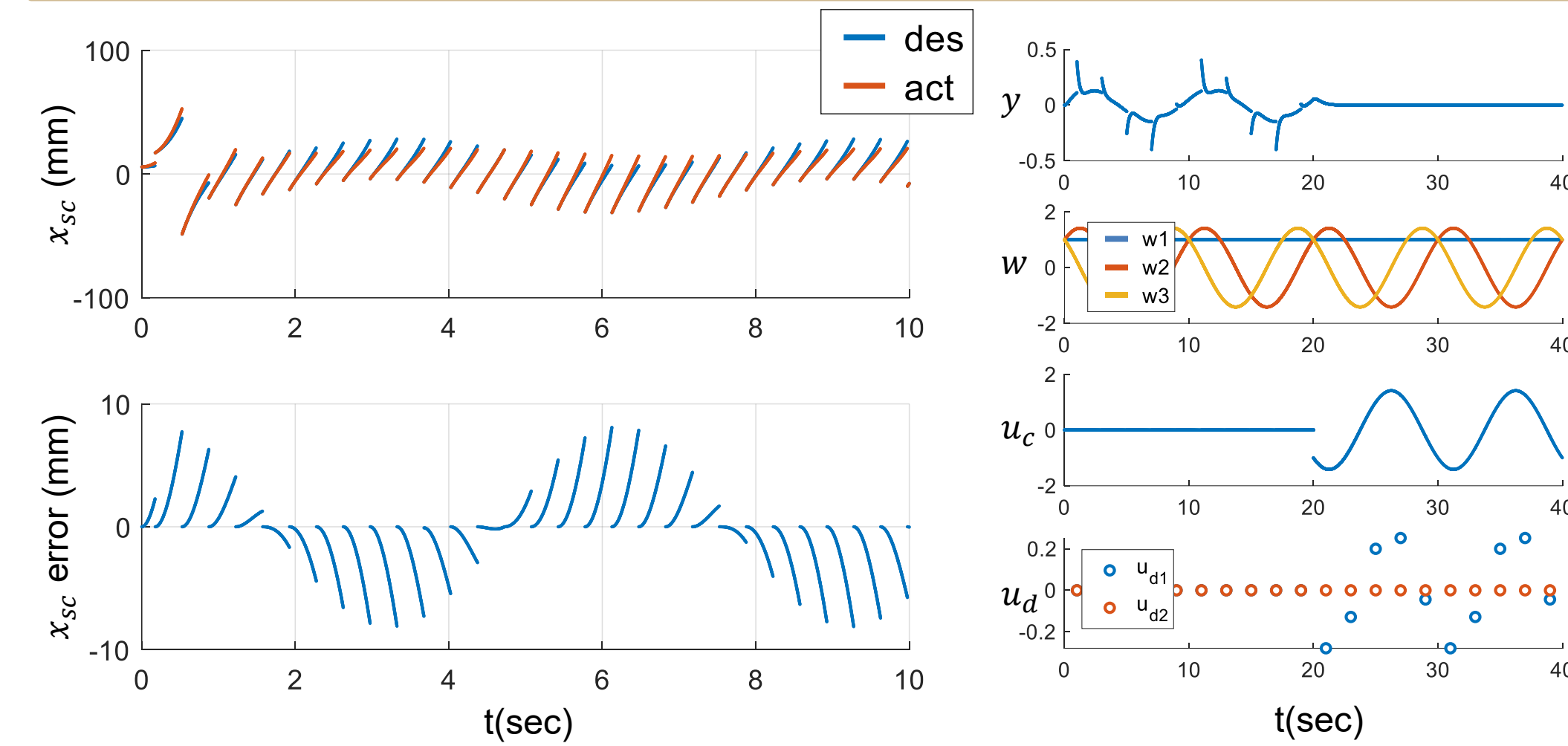
Motivated by bipedal walking on a moving platform, this work studies the disturbance rejection problem of a hybrid linear system subjected to sinusoidal disturbances with unknown frequencies. We propose an adaptive output feedback controller that can learn the correct controller parameters online through a modified recursive least-squares adaptive law to reject the disturbance. Simulation and experimental results show that with the proposed adaptive controller, the center of mass tracking error of a bipedal robot walking on a moving platform with unknown motion is reduced.

Motivation



(left) Bipedal robot walking on a moving ground, e.g., a swaying treadmill. (mid) A linear inverted pendulum model on a moving platform is used to capture the overall dynamic behavior of the robot. (right) Illustration of the hybrid nature of locomotion when taking a step.

The error system is **hybrid** and affected by **unknown time-varying disturbance**.



Problem Formulation

Consider the hybrid dynamical system

$$\begin{aligned} \begin{bmatrix} \dot{x}(t) \\ \dot{w}(t) \end{bmatrix} &= \begin{bmatrix} A_c & P_c \\ 0 & S_c \end{bmatrix} \begin{bmatrix} x(t) \\ w(t) \end{bmatrix} + \begin{bmatrix} B_c \\ 0 \end{bmatrix} u_c(t), & t \neq t_k \\ \begin{bmatrix} x(t_k^+) \\ w(t_k^+) \end{bmatrix} &= \begin{bmatrix} A_d & P_d \\ 0 & S_d \end{bmatrix} \begin{bmatrix} x(t_k^-) \\ w(t_k^-) \end{bmatrix} + \begin{bmatrix} B_d \\ 0 \end{bmatrix} u_d(t), & t = t_k \\ y(t) &= Cx(t) + Qw(t) \end{aligned}$$

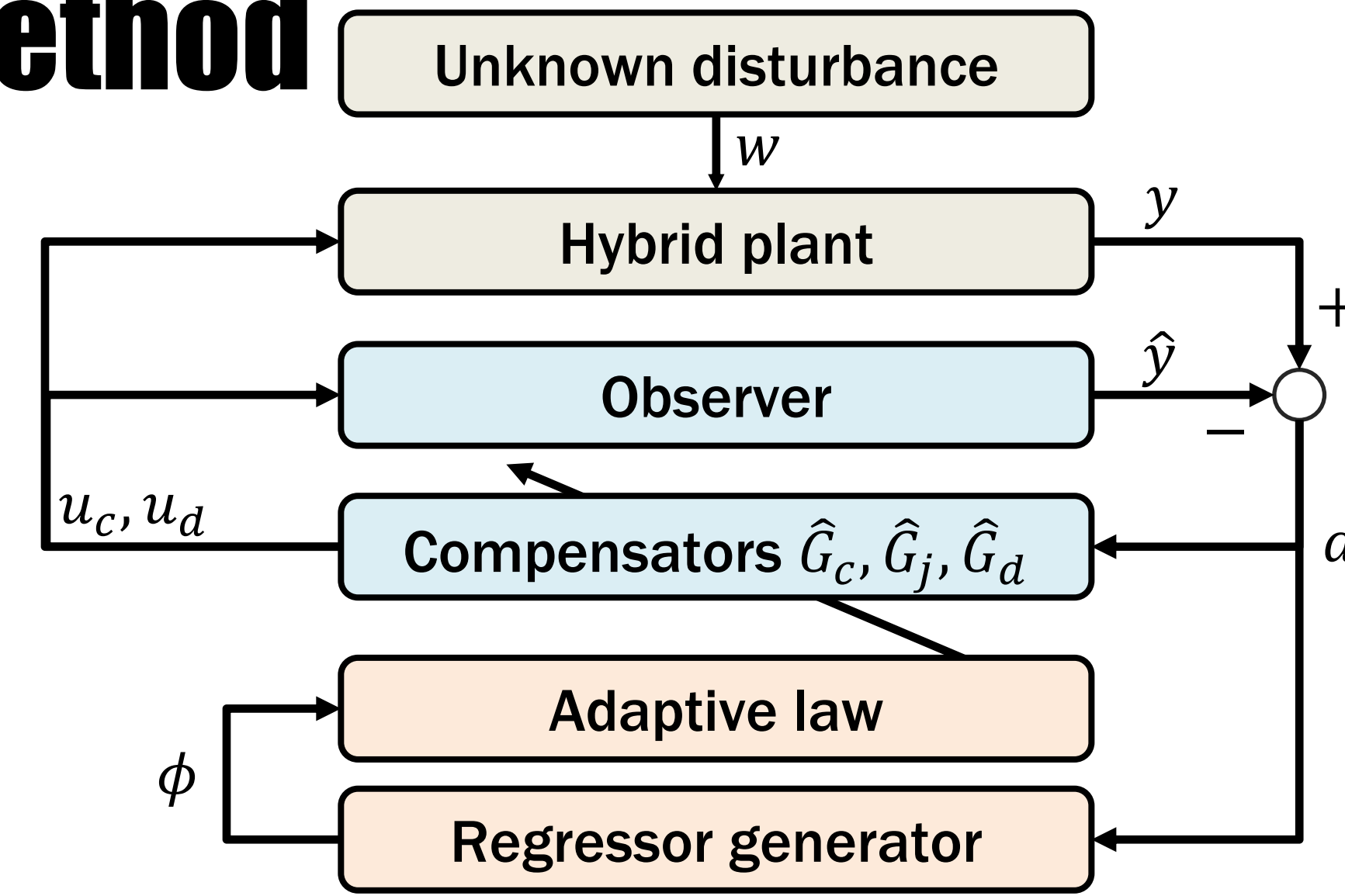
where $t_{k+1} - t_k = T$ for all $k \in \mathbb{N}$, the plant is stable, the disturbance may be modeled as a linear combination of $n_\omega > 0$ sinusoidal components with phase jumps across the discrete events, i.e.,

$$\begin{aligned} S_c &= \text{diag} \left(0, \omega_1 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \dots, \omega_{n_\omega} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right) \\ S_d &= \text{diag} \left(1, \begin{bmatrix} \cos \varphi_1 & \sin \varphi_1 \\ -\sin \varphi_1 & \cos \varphi_1 \end{bmatrix}, \dots, \begin{bmatrix} \cos \varphi_{n_\omega} & \sin \varphi_{n_\omega} \\ -\sin \varphi_{n_\omega} & \cos \varphi_{n_\omega} \end{bmatrix} \right) \end{aligned}$$

where, for $i = 1, 2, \dots, n_\omega$, $\omega_i > 0$ are the distinct disturbance frequencies, and $\varphi_i \in \mathbb{R}$ are the phase jumps across the discrete events.

Assuming A_c, A_d, B_c, B_d, C, T are known and S_c, S_d, P_c, P_d, Q are unknown, the control objective is to design a controller which generates the control input $u_c(t)$ and $u_d(t_k^-)$ such that the measured output $y \rightarrow 0$ as $t \rightarrow \infty$

Method



Step 1: Is it solvable?

Theorem 1 (Hybrid Regulator Equations): For any P_c, P_d and Q , there exists a control input $u_c = \Psi_c w, u_d = \Psi_d w$ such that the measured output $y \rightarrow 0$ as $t \rightarrow \infty$ if and only if there exists a solution (Π_c, Ψ_c, Ψ_d) to the Hybrid Regulator Equations:

$$\begin{aligned} \Pi_c S_c &= A_c \Pi_c + B_c \Psi_c + P_c \\ C \Pi_c + Q &= 0 \\ A_d \Pi_c + B_d \Psi_d + P_d - \Pi_c S_d &= 0 \end{aligned}$$

Step 2: A controller that is safe to adapt

Observer predicts the behavior of the plant without the effect of disturbance

$$\begin{aligned} \hat{\dot{x}}(t) &= A_c \hat{x}(t) + B_c u_c(t), & t \neq t_k \\ \hat{x}(t_k^+) &= A_d \hat{x}(t_k^-) + B_d u_d(t_k^-), & t = t_k \\ \hat{y}(t) &= C \hat{x}(t) \\ d(t) &= y - \hat{y}(t) \end{aligned}$$

Jump compensator produces the discrete control $u_d(t_k^-)$ from $d(t_k^-)$ by

$$\begin{aligned} \eta_j &= 0, & t \neq t_k \\ \eta_j(t_k^+) &= F_j \eta_j(t_k^-) + G_j d(t_k^-), & t = t_k \\ u_d(t_k^-) &= H_j \eta_j \end{aligned}$$

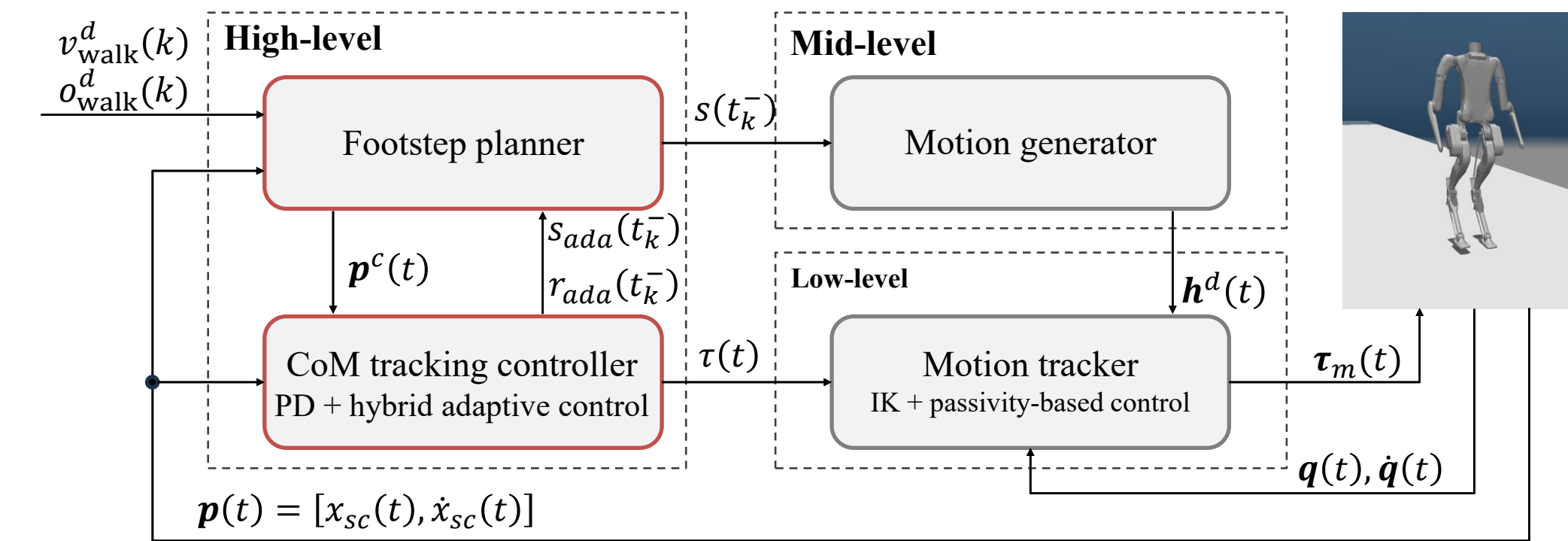
Continuous compensator produces the continuous control input $u_c(t)$ from the signal $d(t)$ by

$$\begin{aligned} \begin{bmatrix} \dot{\eta}_c \\ \dot{\eta}_t \end{bmatrix} &= \begin{bmatrix} F_c & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \eta_c \\ \eta_t \end{bmatrix} + \begin{bmatrix} G_c \\ 0 \end{bmatrix} d(t), & t \neq t_k \\ \begin{bmatrix} \eta_c(t_k^+) \\ \eta_t(t_k^+) \end{bmatrix} &= \begin{bmatrix} I & H_t \\ 0 & F_t \end{bmatrix} \begin{bmatrix} \eta_c(t_k^-) \\ \eta_t(t_k^-) \end{bmatrix} + \begin{bmatrix} 0 \\ G_t \end{bmatrix} d(t_k^-), & t = t_k \\ u_c(t) &= [H_c \quad 0] \begin{bmatrix} \eta_c \\ \eta_t \end{bmatrix} \end{aligned}$$

Theorem 2 (Performance of the controller): Assume there exists a solution to the Hybrid Regulator Equations. There exists compensator parameters (G_j, G_c, G_t) such that the steady-state output of the controller is $u_c(t) = \Psi_c w(t)$ and $u_d(t_k^-) = \Psi_d w(t_k^-)$ and the steady-state measured output is $y(t) = 0$ if $n_j \geq n_w$, $n_c \geq n_x + n_w$, $n_t \geq n_w$ and the pairs $(S_d \exp(S_c T), Q_e)$ and $(\text{diag}(A_c, S_c), [C \quad Q_0])$ are observable.

Bipedal walking

The hybrid adaptive controller is integrated into a hierarchical locomotion control framework for a bipedal robot.



The system of CoM tracking error ε :

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \varepsilon \\ \dot{\varepsilon} \end{bmatrix} &= \left\{ \begin{bmatrix} 0 & 1 \\ -k_p & -k_d \end{bmatrix} + \Delta \dot{z}_{ws}(t) \right\} \begin{bmatrix} \varepsilon \\ \dot{\varepsilon} \end{bmatrix} + \begin{bmatrix} 0 \\ \bar{d}(t) + \tau_{ada} \end{bmatrix} \\ \begin{bmatrix} \varepsilon(t_k^+) \\ \dot{\varepsilon}(t_k^+) \end{bmatrix} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \varepsilon(t_k^-) \\ \dot{\varepsilon}(t_k^-) \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} s_{ada}(t_k^-) \\ r_{ada}(t_k^-) \end{bmatrix} \end{aligned}$$

Step 3: Online parameter estimation

The observer-compensator subsystem:

$$\dot{z}(t) = \bar{A}_c z(t) + \bar{B}_c \Xi_c^T(t) \hat{\theta}_c(t), \quad t \neq t_k$$

$$z(t_k^+) = \bar{A}_d z(t_k^-) + \bar{B}_d \Xi_d^T(t_k^-) \begin{bmatrix} \hat{\theta}_j(t_k^-) \\ \hat{\theta}_d(t_k^-) \end{bmatrix}, \quad t = t_k$$

$$\hat{y}(t) = \bar{C} z(t)$$

where $(\hat{\theta}_c, \hat{\theta}_j, \hat{\theta}_d)$ are the vectorized estimates $(\hat{G}_c, \hat{G}_j, \hat{G}_d)$ and (Ξ_c^T, Ξ_d^T) rearrange $d(t)$.

Lemma 1 (Hybrid Swapping Lemma): the trajectory of $\hat{y}$ is

$$\hat{y}(t) = \varepsilon(t) + \phi_c^T(t) \hat{\theta}_c(t) + [\phi_j^T(t), \phi_d^T(t)] \begin{bmatrix} \hat{\theta}_j(t_k^-) \\ \hat{\theta}_d(t_k^-) \end{bmatrix}$$

where $\varepsilon(t)$ contains the initial condition effect and swapping terms and $\phi_c^T, \phi_j^T, \phi_d^T$ can be generated by a known hybrid system with $d(t)$ as input.

The Hybrid Swapping Lemma yields the construction of a **linear parameter estimation equation**:

$$d(t) = \phi^T(t) \theta^*$$

where θ^* contains the parameters $\theta_c^*, \theta_j^*, \theta_d^*$ that reject the disturbance and $\phi^T(t) = -[\phi_c^T(t), \phi_j^T(t), \phi_d^T(t)]$.

Using **modified recursive least-squares adaptive law** with normalized estimation error, the parameter can be updated.

$$\epsilon = \frac{d - \phi^T \hat{\theta}}{1 + \text{trace}(\phi^T K \phi)}, \quad \hat{\theta} = \text{proj}(\alpha K \phi \epsilon)$$

The covariance matrix update law

$$\dot{K} = -\alpha \frac{K \phi \phi^T K}{1 + \text{trace}(\phi^T K \phi)} + \beta I + \gamma K - \delta K^2$$

Theorem 3: Then all signals in the closed-loop system are uniformly bounded and the measured output converges to zero $\lim_{t \rightarrow \infty} y(t) = 0$, with no knowledge of the disturbance.



Full Paper



Exp Video

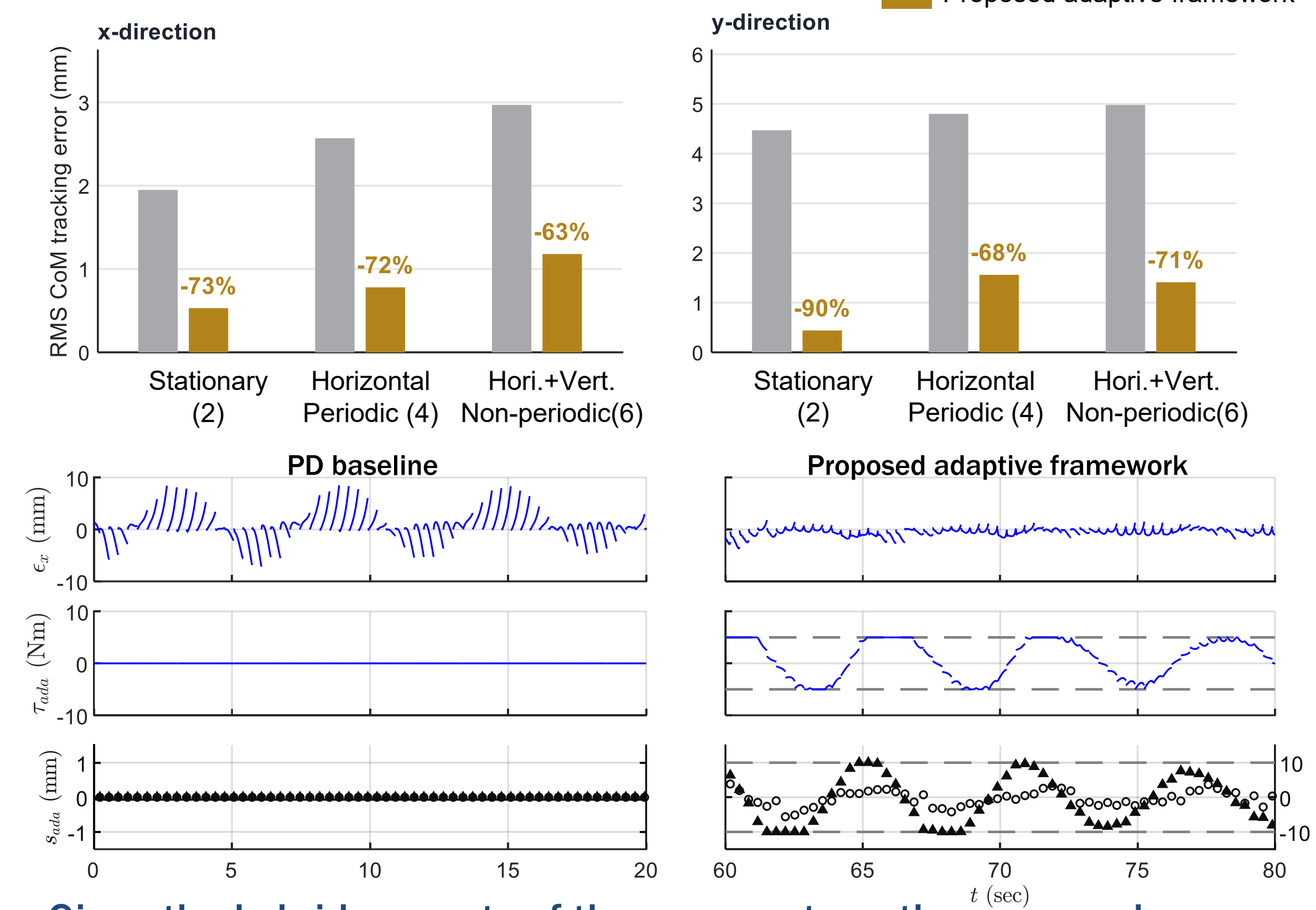


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Simulation (MuJoCo)

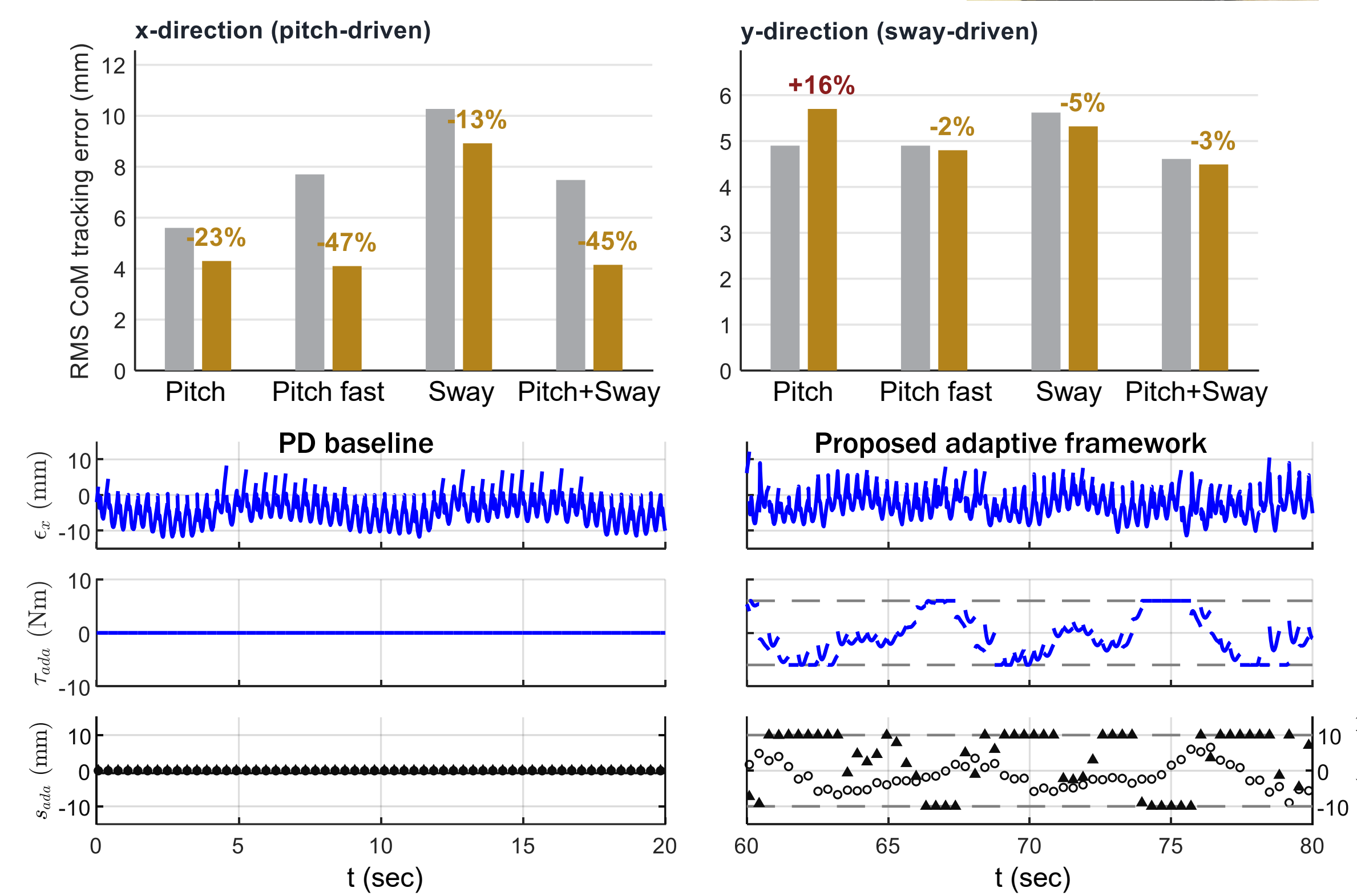
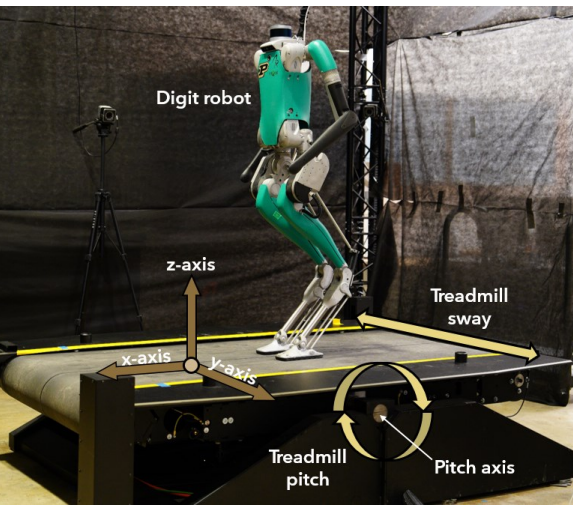
12 unknown ground motions: horizontal, vertical & non-periodic motion. 2 speeds (0, 0.15 m/s).



Given the hybrid property of the error system, the proposed controller can reject the disturbance.

Sim-to-Real results

Treadmill pitches and sways; motion unknown to the robot. 4 profiles.



Sim-to-Real challenges

Unknown time-varying disturbance in non-inertial environments [1]	The proposed adaptive framework for hybrid linear systems
High-frequency, unmodeled disturbance	Low-pass compensators design
Parameter estimation error	Robust adaptive law

Conclusion

We developed an adaptive controller that can reject the sinusoidal disturbances with unknown frequencies, phases, and amplitudes for a class of hybrid linear systems. Then, applying the proposed controller to control a humanoid walking on moving ground improves the center of mass tracking performance.

Reference

- [1] I.C. Chang, X. Huang, T.Y. Lin, S. Teng, W. Li, M. Ghaffari, J. Yi, and Y. Gu, "A survey of legged robotics in non-inertial environments: Past, present, and future," arXiv preprint arXiv:2604.20990, 2026.
- [2] P. Ioannou and B. Fidan, Adaptive Control Tutorial. Philadelphia, PA: Society for Industrial and Applied Mathematics, 2006.
- [3] J. Stewart, I.-C. Chang, Y. Gu, and P. A. Ioannou, "Adaptive ankle torque control for bipedal humanoid walking on surfaces with unknown horizontal and vertical motion," in American Control Conference, IEEE, 2025, pp. 4647–4652.